higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE

MATHEMATICS N5

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QUESTION 1

1.1 $$y=\lim_{x\to 0}\frac{\tan x-x}{x-\sin x}\ \frac{0}{0}$$

$$¿\lim_{x\to 0}\frac{sec^2 x-1}{1-\cos x}\ \frac{0}{0}$$

$$¿\lim_{x\to 0}\frac{sec\,x(sec\,x\tan x)}{\sin x}$$

$$¿\lim_{x\to 0}\frac{2}{\cos^2 x}\times\frac{1}{\sin x}\times\frac{\sin x}{\cos x}$$

$$¿\lim_{x\to 0}\frac{2}{\cos^3 x}$$

$$¿2$$ (4)

1.2 $$f(x)=\frac{3}{e^{x^2}-e^x}$$

$$e^{x^2}-e^x=0$$

$$x^2=x$$

$$x(x-1)=0$$

$$x=0\vee x=1$$ (2)

[6]

QUESTION 2

2.1 $$f(x)=\frac{4x-3}{x+2}$$ (4)

$$f'(x)=\lim_{h\to 0}\frac{\frac{4(x+h)-3}{(x+h)+2}-\frac{4x-3}{x+2}}{h}$$

$$f'(x)=\lim_{h\to 0}\frac{\frac{(4x+4h-3)(x+2)-(4x-3)(x+h+2)}{(x+h+2)(x+2)}}{h}$$

$$¿\lim_{h\to 0}\frac{\frac{4x^2+5x+4xh+8h-6-(4x^2+4xh+5x-3h-6)}{(x+h+2)(x+2)}}{h}$$

$$¿\lim_{h \to 0} \frac{\frac{11h}{(x+h+2)(x+2)}}{h} = \frac{11}{(x+2)^2}$$

2.2 2.2.1 $y=\ln\left(\frac{2x+1}{2x-1}\right)+2\tan^{-1}(2x)$

$$\frac{dy}{dx}=\frac{2x-1}{2x+1}\times\frac{2(2x-1)-2(2x+1)}{(2x-1)^2}+2\times\frac{1}{1+(2x)^2}\times 2$$ (5)

2.2.2 $y=\left\{\sin\left[\tan\left(\cos x^3\right)\right]\right\}^5$

$$\frac{dy}{dx}=5\left\{\sin\left[\tan\left(\cos x^3\right)\right]\right\}^4\times\cos\left[\tan\left(\cos x^3\right)\right]\times sec^2\left(\cos x^3\right)\times\left(-\sin x^3\right)\times 3x^2$$ (5)

2.3 $y=(2x+1)^{\sin 4\pi x}$

$$\ln y=\sin 4\pi x\ln(2x+1)$$

$$\frac{1}{y}\frac{dy}{dx}=4\pi\cos 4\pi x\times\ln(2x+1)+\sin 4\pi x\times\frac{1}{2x+1}\times 2$$

$$\frac{dy}{dx}=y\left[4\pi\cos 4\pi x\ln(2x+1)+\frac{2\sin 4\pi x}{2x+1}\right]$$

$$¿(2x+1)^{\sin 4\pi x}\left[4\pi\cos 4\pi x\ln(2x+1)+\frac{2\sin 4\pi x}{2x+1}\right]$$ (4)

2.4 $cosec^{-1}x+\cot^{-1}(x+y)=\pi$

$$\frac{-1}{x\sqrt{x^2-1}}-\frac{1}{1+(x+y)^2}\left(1+\frac{dy}{dx}\right)=0$$

$$\frac{-1}{1+(x+y)^2}\left(1+\frac{dy}{dx}\right)=\frac{1}{x\sqrt{x^2-1}}$$

$$1+\frac{dy}{dx}=\frac{-1+(x+y)^2}{x\sqrt{x^2-1}}$$

$$\frac{dy}{dx}=\frac{-1+(x+y)^2}{x\sqrt{x^2-1}}-1$$ (4)
[22]

QUESTION 3

3.1 3.1.1 $f(x)=x^3-2x+3$

$f'(x)=3x^2-2=0$

$x=\pm\sqrt{\frac{2}{3}}$

$y=1{,}911\ y=4{,}089$

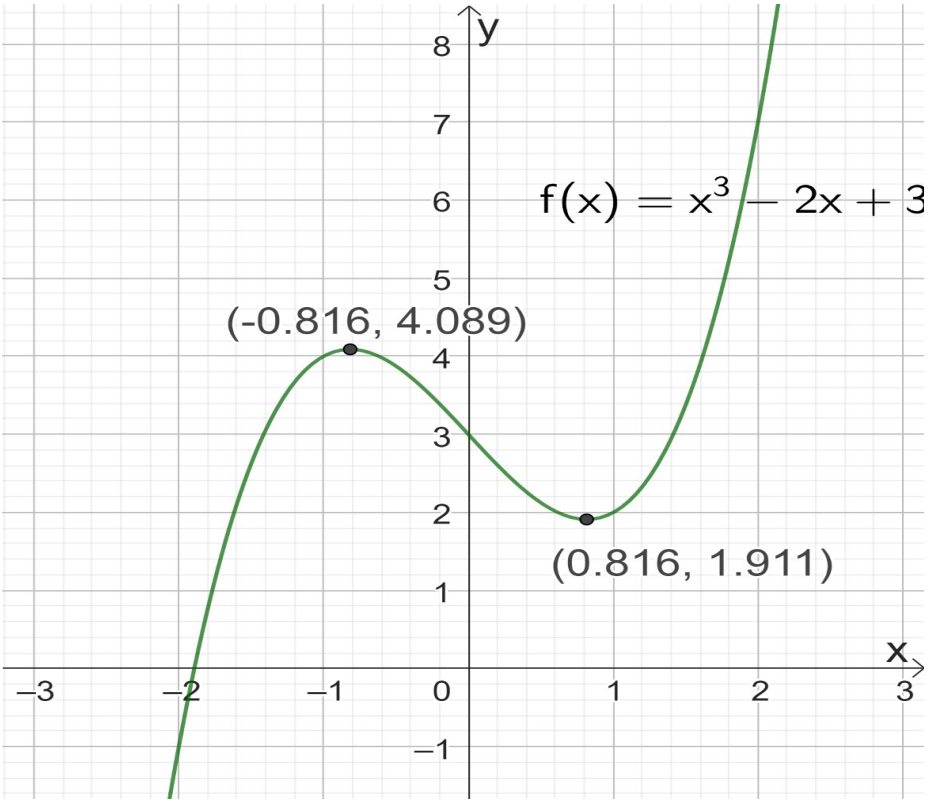
Thus, the coordinates of the turning points are $\left(-\sqrt{\frac{2}{3}};4{,}089\right)$ and $\left(\sqrt{\frac{2}{3}};1{,}911\right)$ (3)

3.1.2

x	-2	-1	0	1	2
y	-1	4	3	2	7

= 1 mark for any 2 correct answers (2)

3.1.3



=1 mark for the shape
= 1 mark for the indication of the turnings point on the graph. (2)

3.1.4 $Let\ x_0 = -1,75$

$f(-1,75) = 1,141$

$f'(-1,75) = 7,188$

$x_1 = -1,75 + \frac{1,141}{7,188}$

¿ $-1,909$

$f(-1,909) = -0,136$

$f'(-1,909) = 8,929$

$x_2 = -1,909 - \frac{0,136}{8,929}$

¿ $-1,893$ (4)

3.2 Let $x =$ ¿ the width of the pen.
$y =$ ¿ the side of each of the smaller rectangle pens

$P = 4x + 6y$

$120 = 4x + 6y$

$y = \frac{120 - 4x}{6}$

$A = x \times 3y$

¿ $x \times 3\left(\frac{120 - 4x}{6}\right)$

¿ $60x - 2x^2$

$\frac{dA}{dx} = 60 - 4x = 0$

$x = 15\ y = 10$

Thus, for maximum area width¿ 15 and length ¿ 30

$Area = 450\,m^2$ (6)

3.3 $C=2\pi r$

$$\frac{dC}{dt}=2\pi\frac{dr}{dt}$$

$$\frac{dA}{dt}=2\pi(-4\,cm)$$

$$\therefore -8\pi\,cm/s=-25{,}133\,cm/s$$ (3)

[20]

QUESTION 4

4.1 4.1.1 $\int \frac{\cos x}{2\sin x-1}dx$

$let\ u=2\sin x-1$

$du=2\cos x\,dx$

$$\therefore \frac{1}{2}\int \frac{1}{u}du$$

$$\therefore \frac{1}{2}\ln u+C$$

$$\therefore \frac{1}{2}\ln(2\sin x-1)+C$$ (3)

Alternative

$let\ f(x)=2\sin x-1$

$f'(x)=2\cos x$

$$\int \frac{\cos x}{2\sin x-1}dx=\frac{1}{2}\int \frac{2\cos x}{2\sin x-1}dx$$

$$\therefore \frac{1}{2}\ln(2\sin x-1)+C$$

4.1.2 $\int \sin\left(\frac{7}{3}x\right)\sin\left(\frac{1}{4}x\right)dx$

$$\therefore \frac{1}{2}\int\left[\cos\left(\frac{25}{12}x\right)-\cos\left(\frac{31}{12}x\right)\right]dx$$

$$\therefore \frac{6}{25}\sin\left(\frac{25}{12}x\right)-\frac{6}{31}\sin\left(\frac{31}{12}x\right)+C$$ (3)

4.1.3 $y=\frac{e^x}{\sqrt{3-e^{2x}}}$

$\therefore \int \frac{e^x}{\sqrt{3-e^{2x}}}dx$

$let\ u=e^x \therefore du=e^x dx \therefore dx=\frac{du}{e^x} \wedge e^{2x}=u^2$

$\therefore \int \frac{1}{\sqrt{(\sqrt{3})^2-u^2}}du$

$\therefore \sin^{-1}\frac{u}{\sqrt{3}}+c$

$\therefore \sin^{-1}\frac{e^x}{\sqrt{3}}+c$ (4)

4.1.4 $\int \frac{2x^3+8x^2-6x+10}{x-2}dx$

$\therefore \int \left(2x^2+12x+18+\frac{46}{x-2}\right)dx$

$\therefore \frac{2}{3}x^3+6x^2+18x+46\ln(x-2)+C$ (5)

4.1.5 $\int \log_2 x\,dx$

$let\ u=\log_2 x\ dv=dx$ OR $f(x)=\log_2 x\ g'(x)=1$

$du=\frac{1}{x\ln 2}dx\ v=xf'(x)=\frac{1}{x\ln 2}g(x)=x$

$\int \log_2 x\,dx=x\log_2 x-\int \frac{1}{x\ln 2}\times x dx$

$\therefore x\log_2 x-\frac{1}{\ln 2}x+C$ (4)

4.2 $v=\int \frac{dv}{1-(kv)^2}$

$\frac{1}{(1+kv)(1-kv)}=\frac{A}{1+kv}+\frac{B}{1-kv}$

$1=A(1-kv)+B(1+kv)$

$let\ v=\frac{1}{k};A=\frac{1}{2}$

¿

$v=\frac{-1}{k};B=\frac{1}{2}$

¿ $\frac{1}{2}\int\left(\frac{1}{1+kv}+\frac{1}{1-kv}\right)dv$

¿ $\frac{1}{2k}\left[\ln(1+kv)-\ln(1-kv)\right]+C$

¿ $\frac{1}{2k}\ln\left(\frac{1+kv}{1-kv}\right)+C$ (5)

[24]

QUESTION 5

5.1 $\int_1^{\infty}\frac{1}{x^3}dx$

¿ $\lim_{b\to\infty}\int_1^{b}\frac{1}{x^3}dx$

¿ $\lim_{b\to\infty}\left[\frac{-1}{2x^2}\right]_1^{b}$

¿ $\lim_{b\to\infty}\left[\frac{-1}{2b^2}+\frac{1}{2}\right]$

¿ $\frac{1}{2}$

May accept direct working without $\lim_{b\to\infty}$ ¿

$\int_1^{\infty}\frac{1}{x^3}dx$

¿ $\left[\frac{-1}{2x^2}\right]_1^{\infty}$

¿ $\left[\frac{-1}{\infty}+\frac{1}{2}\right]$

¿ $\frac{1}{2}$ (4)

5.2 5.2.1 $(x-2)^2=-x^2+2x$

$2x^2-6x+4=0$

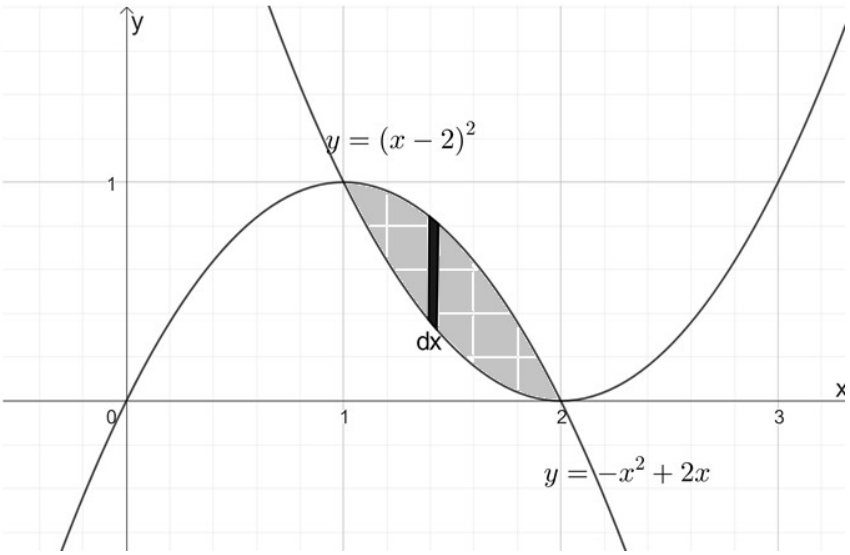
$x^2-3x+2=0$

$(x-1)(x-2)=0$

$x=1 \vee x=2$
$y=1\ y=0$

Thus, the coordinates of the points of intersection for bounded area are $(1;1)$ and $(2;0)$. (3)

5.2.2



= 1 mark for indication of the correct enclosed area
= 1 mark for indication of the vertical or horizontal strip (2)

5.2.3 $A=\int_a^b (y_1-y_2)\,dx$

$\int_1^2 \left[(-x^2+2x)-(x-2)^2\right]dx$

$\int_1^2 (-2x^2+6x-4)\,dx$

$\left[\frac{-3}{2}x^2+3x^2-4x\right]_1^2$

$\frac{1}{3}\,units^2 \vee 0{,}333\,units^2$ (4)

5.2.4 $V=\pi\int_a^b\left(y_1{}^2-y_2{}^2\right)dx$

$\therefore\pi\int_1^2\left[\left(-x^2+2x\right)^2-\left(x^2-4x+4\right)^2\right]dx$

$\therefore\pi\int_1^2\left(4x^3-20x^2+32x-16\right)dx$

$\therefore\pi\left[x^4-\frac{20}{3}x^3+16x^2-16x\right]_1^2$

$\therefore\frac{\pi}{3}units^3\vee\therefore 1{,}047\,units^3$ (5)

5.3 $d)_z=r^2\,dA$

$\therefore x^2\cdot 2\pi x\,dx$

$I_z=2\pi\int_0^1 x^3\,dx$

$\therefore 2\pi\left[\frac{x^4}{4}\right]_0^1$

$\therefore\frac{\pi}{2}cm^4\vee\therefore 1{,}571\,cm^4$ (4)
[22]

QUESTION 6

6.1 $\frac{dy}{dx}=\frac{3x(y-5)}{x^2-1}$

$\frac{dy}{y-5}=\frac{3x}{x^2-1}dx$

$\ln(y-5)=\frac{3}{2}\ln\left(x^2-1\right)+C$ (3)

6.2 $\frac{d^2y}{dx^2} = \sin^2 3x$

$\frac{d^2y}{dx^2} = \frac{1}{2} - \frac{1}{2}\cos 6x$

$\frac{dy}{dx} = \frac{1}{2}x - \frac{1}{12}\sin 6x + A$

$y = \frac{1}{4}x^2 + \frac{1}{72}\cos 6x + Ax + B$ (3)

[6]

TOTAL: 100

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